Academic Methodologies Paper

I. Drews[1]

[1]Film University Babelsberg KONRAD WOLF, Germany

Abstract

Here, I will state the problem I am trying to solve, which is to determine how AI-supported curatorial tools influence museum visitors' engagement and whether they perform better or worse than traditional curation in helping audiences discover, access, and connect with museum archives. Then I will explain why this is both interesting and important and summarize some of my solutions.

1. Introduction

Artificial intelligence is increasingly shaping the ways individuals interact with their environments, influencing everything from transportation to cultural experiences. In particular, museums have been integrating machine learning models into their curatorial and visitor engagement strategies, showing a shift in how visitors are engaging with the archive. This paper will be exploring how machine learning models have been changing the museum visitors experience and the different curatorial methods they enable in comparison to traditional curatorial methods. This will be done using the case study XCurator, an AI tool made to enrich museum visitors' experiences through the use of various AI models. Through this comparison, the paper will explore which engagement strategies are most effective and will consider the broader implications of curatorial strategies. Ultimately, this exploration aims to offer museums actionable insights into how AI tools can enhance and hinder visitor engagement.

2. Literature Review: Machine learning models in curatorial practice

In this section, I thought of first covering traditional curatorial practices, then looking at the emerging uses of AI in museums, giving a short discussion of visitor engagement theory, and lastly, looking at some related work.

3. Method: A Comparative Case Study Approach

Explanation of how I will be doing the comparative analysis of traditional curatorial methods vs. AI museum apps. I will end this section acknowledging some of the limitations of this methodology.

3.1. Case Study: XCurator / Gropius Bau's Prisma

I will explain the origins and objectives of the app I end up choosing, some of its technical architecture, as well as the AI models it uses. I will also explain how they implemented it into the museum environment, visitor interaction, and lastly, some of the challenges and opportunities observed.

3.2. Comparative Analysis: Traditional vs. AI-Supported Museum Practices

Here I will explain some of the observations I find, such as visitor engagement outcomes, curatorial creativity and flexibility, accessibility, and some potential biases and unintended consequences. The things I look for will be shaped by the readings I choose, so these will probably change.

3.3. Discussion: Implications for museum curation

I will discuss some of the implications these apps will have for museum curation and the shift in the roles of curators and visitors. Once again, this will probably change according to what I decide to focus on.

3.4. References

The three references I have so far are:

1. [Lee22] 2. [SBB*24] 3. [TB23]

4. Conclusion

I will give a summary of my key findings, some of the contributions I have for museum studies and curatorial practices in terms of integrating these apps, and lastly, ideas for future research.

References

[Lee22] Lee B. C. G.: The "Collections as ML Data" Checklist for Machine Learning & Cultural Heritage, July 2022. arXiv:2207.02960, doi:10.48550/arXiv.2207.02960. 1

[SBB*24] SCHAERF L., BALLESTEROS P., BERNASCONI V., NERI I., DEL CASTILLO D. N.: AI Art Curation: Re-imagining the city of Helsinki in occasion of its Biennial, Oct. 2024. arXiv:2306.03753, doi:10.48550/arXiv.2306.03753. 1

[TB23] THIEL S., BERNHARDT J. C. (Eds.): *AI in Museums: Reflections, Perspectives and Applications*, 1 ed., vol. 74 of *Edition Museum*. transcript Verlag, Bielefeld, Germany, Dec. 2023. doi:10.14361/9783839467107. 1